Public-data training pilot

AI-guided cellular rejuvenation research

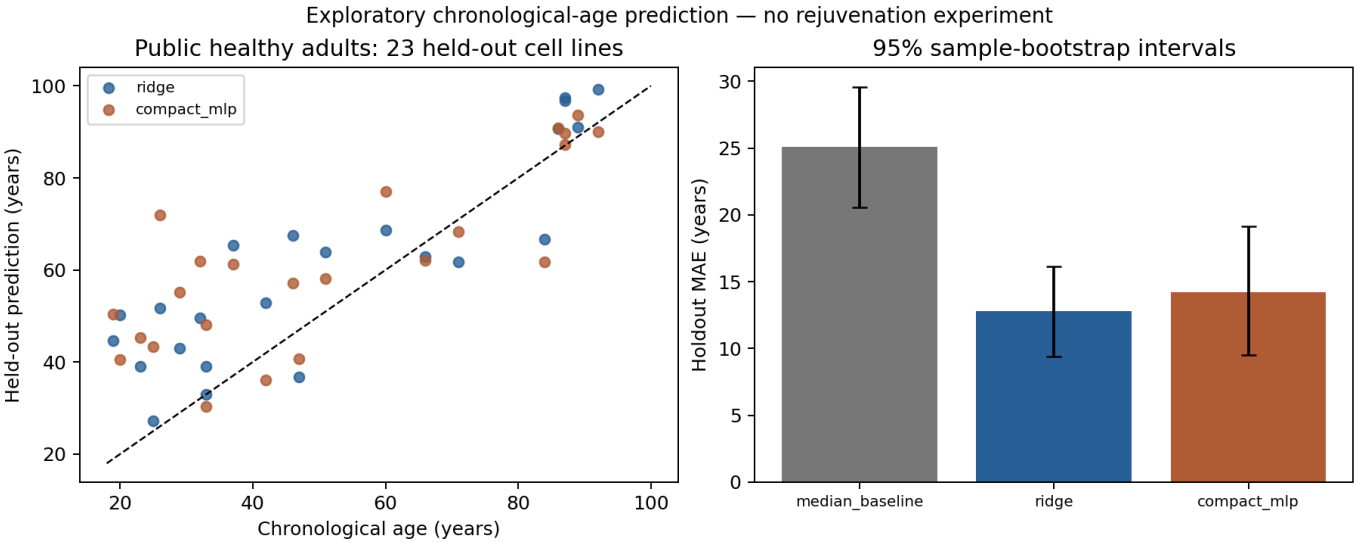
David Xu, PhD; Peter Lee, PhD, MD; and Zhimin Zheng, PhD

A small, reproducible study learns chronological-age associations in cultured adult fibroblasts. It is an inspectable engineering milestone for a proposed research program. It does not show biological age reversal, identify a causal rejuvenation mechanism or validate a therapeutic target.

Free public data: GEO GSE113957, generated by Fleischer et al. (2018). Of 143 samples, 111 healthy adult cell lines qualified; 88 were used for training and 23 for one frozen holdout. All progeria and pediatric samples were excluded. No paid data, raw-sequencing download, GPU or cloud training was used.

Model	Holdout MAE, years	95% bootstrap interval
Training-median baseline	25.07	20.59 to 29.59
Ridge	12.79	9.39 to 16.15
Compact neural network	14.24	9.53 to 19.17

The neural model did not outperform Ridge. Its paired MAE improvement over Ridge was -1.45 years (95% interval -5.05 to 1.98). Both models passed the predeclared comparison with the median baseline. This is not a comparison establishing superiority over published aging clocks. The neural component has 145 weights and biases, plus fitted PCA and preprocessing parameters.



Critical limitations: 23 held-out cell lines; one cohort and one split; no external validation. Training median age was 62.5 versus 46 in the holdout. In held-out adults below 60 (n=14), Ridge MAE was 15.84 years and R-squared was -2.41. The intervals above condition on this split and the fitted models. They do not include uncertainty across new cohorts or new training sets.

What the evidence supports

Training and verification

The protocol was frozen before fitting. Public cell IDs defined groups, and no group crossed the training/holdout boundary. Each training fold performed its own transcript filtering, log transformation, variance selection and scaling. Ridge alpha was chosen from four values by grouped inner cross-validation. The neural architecture was fixed: 16 principal components and eight tanh hidden units, with training-only target scaling.

Thirty shuffled-training-label Ridge controls produced mean holdout MAE 24.08 years (range 18.71 to 30.71). This is a diagnostic, not a confirmatory permutation p-value: model selection was not repeated for each shuffle. No holdout-based search for a favorable seed or neural architecture was performed.

Independent software/numeric verification passed 103 checks and all seven tests. Checks covered downloaded data hashes, labels, exclusions, split and inner-fold membership, fitted preprocessing, checkpoint predictions, metrics and bootstrap intervals. A deliberately corrupted prediction record was rejected. V1 source drift from an example-path fix was detected; V2 reran the unchanged protocol with identical predictions and frozen source.

The release includes code, exact dependencies, acquisition scripts, source hashes, sample manifests, per-sample predictions, model checkpoints and result files. CPU fitting took 2.19 seconds here, excluding download, setup, development and verification; direct cloud spending was $0. This timing is not a claim of end-to-end research cost or expected therapeutic development cost.

Boundaries investors should understand

This is an association study on an existing cohort. Sex, ancestry, passage, culture and platform can confound prediction. Public cell IDs are a donor proxy, not independently confirmed donor identity or unrelatedness. Gene coefficients are model weights, not experimentally validated mechanisms. No cells were treated in this project and no cell-function, safety, lifespan or clinical outcome was measured.

The proposed venture has no established biological efficacy, proprietary validated target, institutional sponsorship, founder role commitments or clinical results. The existing $8 million, 24-month program is illustrative. Public code and data reanalysis alone do not establish commercial demand or defensible therapeutic IP.

Next decision gate

Lock an independent adult fibroblast cohort and expression-harmonization protocol before external evaluation. Assess age-band and batch robustness, then separately test whether intervention-associated predictions track independent function while preserving cell identity. Mechanism claims require prospective perturbation, rescue, independent donors, replication, persistence and safety evidence.

Inspect and reproduce

https://github.com/xuda1979/anti-aging
Study: results/pilot_20261021_v2/REPORT.md
Verification: verification/REPORT.md
Contact: David Xu, PhD - exutechnology@gmail.com

Sources: NCBI GEO GSE113957, https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE113957 ; Fleischer et al., Genome Biology 19, 221 (20 December 2018), https://doi.org/10.1186/s13059-018-1599-6 . Original investigators generated the data. Public download does not establish unrestricted commercial rights. This addendum reports a new limited reanalysis, with AI-assisted implementation and independent computational verification.